Practice Exam One: Version One

Topics Covered:

- KCL and KVL
- Ohm's Law
- Passive Sign Convention
- Power Calculation
- Current and Voltage Division
- Equivalent Resistance of a Simple Resistive Network
- Equivalent Resistance Calculation including Dependent Sources

Practice Exam Tips:

- Treat this as timed practice, but do not rush. The goal is to identify gaps in understanding.
- After completing the exam, you can grade your work using the provided rubric in the solution. The practice solutions also include suggested discussion questions to try.
- If you make mistakes, revisit the relevant topic and rework the problem before reviewing the solution or attempting the second practice exam.
- Focus on method and clarity, not speed.

Exam Tips:

- Show all work clearly. Correct reasoning with minor algebra errors can still earn partial credit.
- Clearly label currents, voltages, and reference directions.
- Use consistent sign conventions throughout each problem.
- Include units in all final answers.
- The actual exam time limit is 75 minutes. If you follow the suggested pacing from this practice exam, you will finish in 60 minutes.

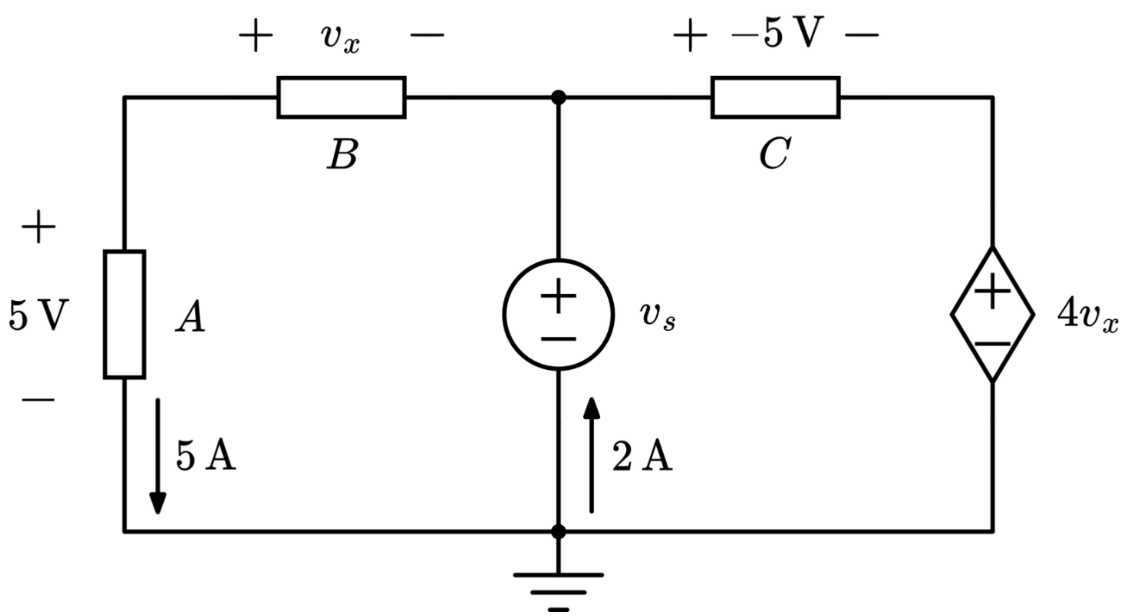
Question One: KCL, KVL, Ohm's Law, and Passive Sign Convention

For the circuit shown below, with three unknown circuit elements, find the following:

a) v_x
b) v_s
c) The resistance of circuit element A

Finally, calculate the power of the voltage-dependent voltage source and state whether the power is absorbed or delivered.

Suggested time limit: 20 minutes

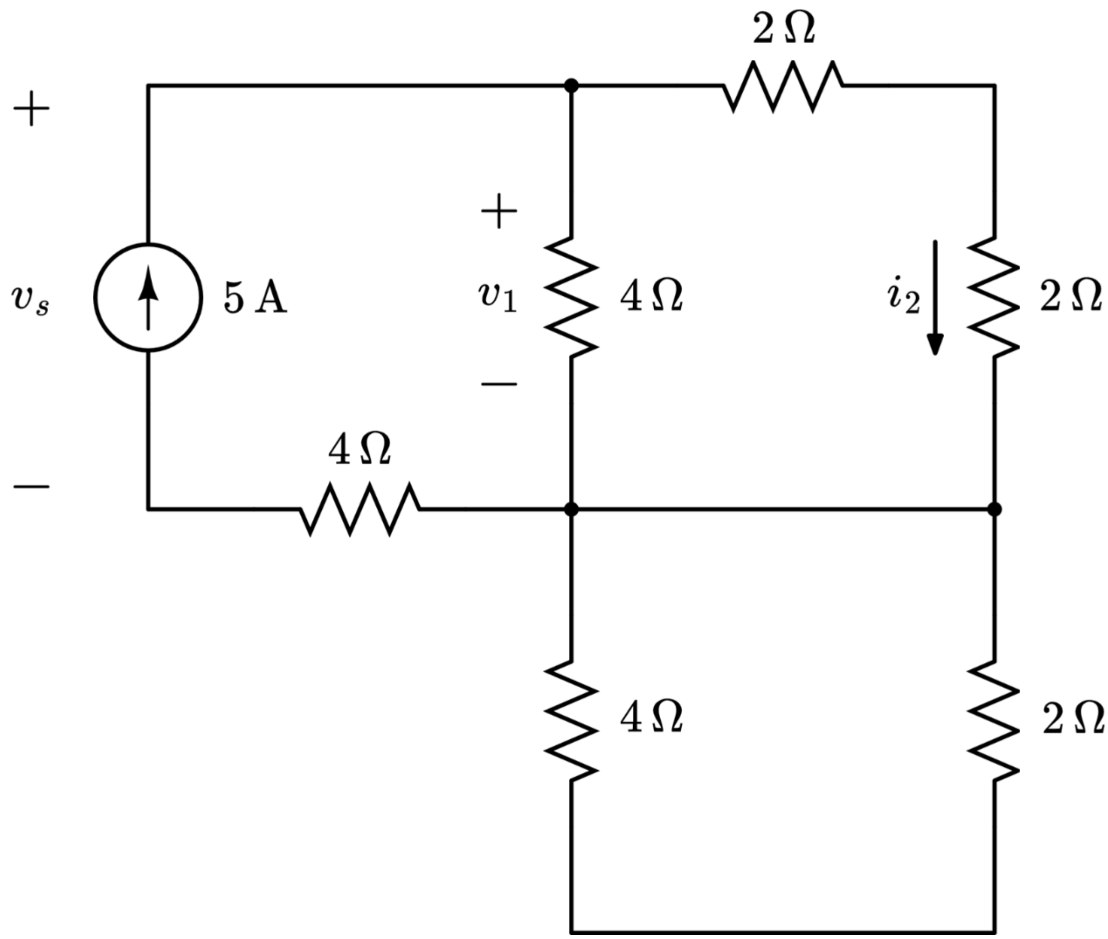


Question Two: Current Division, Voltage Division, and Equivalent Resistance

For the circuit shown below, find the following:

a) v_s
b) v_1
c) i_2

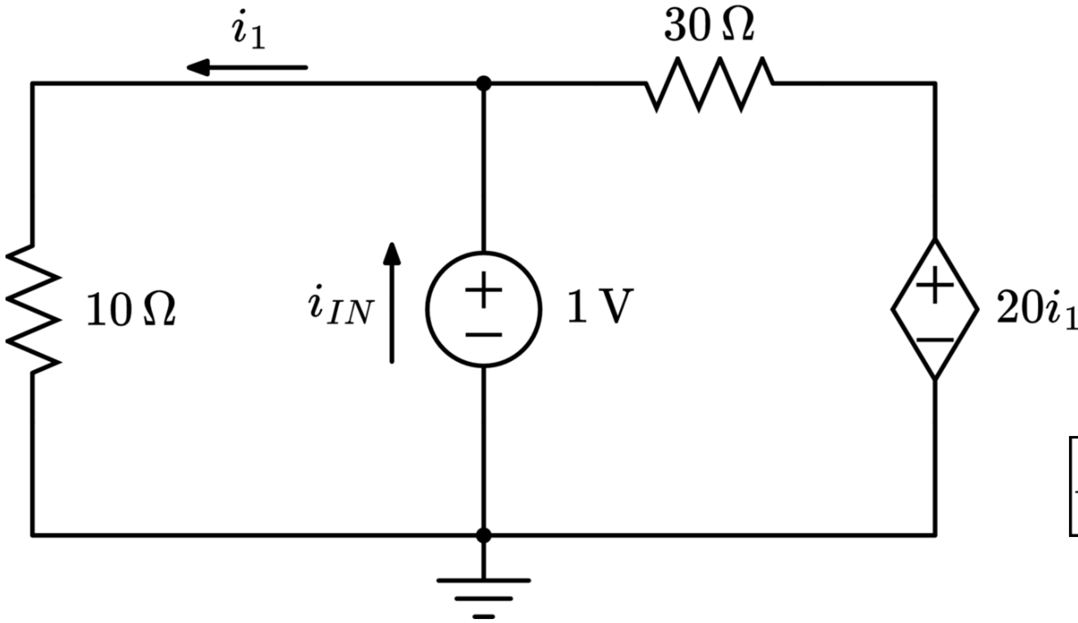
Suggested time limit: 20 minutes



Question Three: Equivalent Resistance with Dependent Sources

For the circuit shown below, find the equivalent resistance seen by the independent voltage source.

Suggested time limit: 20 minutes



$$\boxed{R_{eq} = \frac{v_{IN}}{i_{IN}}}$$